

22050 Signals and Linear Systems in Continuous Time

Formula Collection

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1 Classification of Signals and Systems

1.1 Signal Classification

1.1.1 Continuous-time and discrete-time signals

$$x(t), \quad x[n] = x(nT_s), \quad t \in \mathbb{R}, \quad n \in \mathbb{Z}.$$

A continuous-time signal is defined for all real time values. A discrete-time signal is defined only at sample instants.

1.1.2 Analog, sampled, quantized, and digital signals

$$\Delta V_{\text{LSB}} = \frac{V_{\text{FS}}}{2^N}.$$

Analog signals are continuous in time and amplitude. Digital signals are discrete in time and amplitude, with one least significant bit set by the full-scale range and bit count.

1.1.3 Even and odd signal components

$$x_e(t) = \frac{x(t) + x(-t)}{2}, \quad x_o(t) = \frac{x(t) - x(-t)}{2}, \quad x(t) = x_e(t) + x_o(t).$$

Even components satisfy $x_e(t) = x_e(-t)$. Odd components satisfy $x_o(t) = -x_o(-t)$.

1.1.4 Periodic signal and fundamental period

$$x(t + T_0) = x(t), \quad \omega_0 = \frac{2\pi}{T_0}.$$

The smallest positive T_0 is the fundamental period. Periodic signals have discrete harmonic frequencies $n\omega_0$.

1.1.5 Signal time shift and time scaling

$$y(t) = x(t - t_0), \quad z(t) = x(at).$$

Positive t_0 delays the signal. Time scaling by $|a| > 1$ compresses the signal, and $a < 0$ also reverses time.

1.1.6 Unit step, ramp, and rectangular pulse

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0, \end{cases} \quad r(t) = t u(t), \quad p_T(t) = u(t) - u(t - T).$$

The step gates causal signals. The ramp is the integral of the step, and a rectangular pulse is formed by subtracting delayed steps.

1.1.7 Dirac impulse properties

$$\delta(t) = \delta(-t), \quad \int_{-\infty}^{\infty} \delta(t) dt = 1, \quad \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0).$$

The impulse is an even generalized signal with unit area. Its sampling property extracts the signal value at the impulse location.

1.1.8 Impulse as derivative of step

$$\frac{d}{dt} u(t) = \delta(t), \quad \int_{-\infty}^t \delta(\tau) d\tau = u(t).$$

Impulse and step are related by differentiation and integration. This relation is used when converting between impulse, step, and ramp responses.

1.2 System Classification

1.2.1 Linearity of systems

$$T\{ax_1(t) + bx_2(t)\} = aT\{x_1(t)\} + bT\{x_2(t)\}.$$

A linear system obeys superposition and homogeneity. Terms such as $x^2(t)$, $y^2(t)$, products of signals, and additive constants violate linearity.

1.2.2 Time invariance

$$T\{x(t - t_0)\} = y(t - t_0) \quad \text{when} \quad T\{x(t)\} = y(t).$$

A time-invariant system has unchanged behavior after a time shift. Explicit dependence on t in coefficients usually violates time invariance.

1.2.3 Causality

$$y(t_0) \text{ depends only on } x(t), \quad t \leq t_0.$$

A causal system does not require future input values. Terms such as $x(t + t_0)$ with $t_0 > 0$ are noncausal.

1.2.4 LTIC differential equation

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m x(t)}{dt^m}, \quad a_k, b_m = \text{real constants.}$$

Linear time-invariant continuous-time systems in the course are represented by constant-coefficient differential equations.

1.2.5 Filter type from low-frequency and high-frequency gain

$$H(0), \quad \lim_{\omega \rightarrow \infty} H(j\omega).$$

A lowpass filter has nonzero low-frequency gain and vanishing high-frequency gain. A highpass filter has vanishing low-frequency gain and nonzero high-frequency gain, while a bandpass filter vanishes at both extremes.

2 Time Domain and Impulse Response

2.1 System Equations

2.1.1 Differential operator notation

$$D = \frac{d}{dt}, \quad Q(D)y(t) = P(D)x(t).$$

The operator notation writes constant-coefficient differential equations compactly. $Q(D)$ describes the output-side dynamics and $P(D)$ describes input derivatives.

2.1.2 Characteristic equation

$$Q(s) = a_N s^N + a_{N-1} s^{N-1} + \cdots + a_0 = 0.$$

The roots of $Q(s)$ determine the natural modes of the zero-input response and the poles of the transfer function.

2.1.3 Zero-input and zero-state response

$$y(t) = y_{zi}(t) + y_{zs}(t), \quad y_{zs}(t) = h(t) * x(t).$$

The zero-input response is caused by initial conditions. The zero-state response is caused by the input through the impulse response.

2.1.4 Natural modes from roots

root of $Q(s)$	time-domain mode
p	$Ce^{pt}u(t)$
p repeated r times	$(C_0 + C_1t + \cdots + C_{r-1}t^{r-1})e^{pt}u(t)$
$\sigma \pm j\omega_d$	$e^{\sigma t}(C_1 \cos \omega_d t + C_2 \sin \omega_d t)u(t)$

Real roots give exponential modes. Complex conjugate roots give damped oscillatory modes.

2.2 Impulse, Step, and Ramp Responses

2.2.1 Impulse response definition

$$h(t) = T\{\delta(t)\}.$$

The impulse response fully characterizes an LTIC system in the time domain.

2.2.2 Impulse response from transfer function

$$h(t) = \mathcal{L}^{-1}\{H(s)\}, \quad H(s) = \frac{P(s)}{Q(s)}.$$

With zero initial conditions, the transfer function is the Laplace transform of the impulse response.

2.2.3 Direct feedthrough impulse term

$$H(s) = K + \tilde{H}(s) \iff h(t) = K\delta(t) + \tilde{h}(t).$$

If the numerator order equals the denominator order, polynomial division gives an impulse term. This term represents instantaneous feedthrough.

2.2.4 Step response and impulse response

$$y_{\text{step}}(t) = h(t) * u(t) = \int_{-\infty}^t h(\tau) d\tau, \quad h(t) = \frac{d}{dt}y_{\text{step}}(t).$$

The unit step response is the time integral of the impulse response. Differentiating the step response gives the impulse response, including possible impulse terms.

2.2.5 Ramp response

$$y_{\text{ramp}}(t) = h(t) * r(t) = \int_{-\infty}^t y_{\text{step}}(\tau) d\tau.$$

The unit ramp response is the integral of the step response for causal LTIC systems.

2.3 Stability

2.3.1 BIBO stability from impulse response

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty.$$

Bounded-input bounded-output stability requires an absolutely integrable impulse response.

2.3.2 Pole locations and stability

$$\Re\{p_k\} < 0 \quad \forall k \implies \text{asymptotically stable.}$$

For rational causal LTIC systems, poles in the open left half-plane give decaying natural modes. Poles on the imaginary axis give marginal behavior only if simple, and right-half-plane poles are unstable.

3 Convolution

3.1 Convolution Integral

3.1.1 Continuous-time convolution

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau.$$

Convolution combines an input with an impulse response to produce the zero-state output of an LTIC system.

3.1.2 Causal convolution

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \quad \text{for causal } f, g.$$

When both signals are causal, the integration interval collapses to the overlap between 0 and t .

3.1.3 Convolution properties

$$\begin{aligned} f * g &= g * f, \\ f * (g * h) &= (f * g) * h, \\ f * (g + h) &= f * g + f * h, \\ f * \delta &= f, \quad f * \delta(t - t_0) = f(t - t_0). \end{aligned}$$

Convolution is commutative, associative, and distributive. The impulse is the identity element for convolution.

3.1.4 Differentiation and integration through convolution

$$\frac{d}{dt}(f * g) = \frac{df}{dt} * g = f * \frac{dg}{dt}, \quad u * f = \int_{-\infty}^t f(\tau) d\tau.$$

Differentiation can be applied to either convolution factor. Convolution with the unit step integrates the signal.

3.2 Reusable Convolution Pairs

3.2.1 Exponentials with equal exponents

$$(e^{-at}u(t)) * (e^{-at}u(t)) = te^{-at}u(t).$$

Repeated first-order factors in the Laplace domain produce polynomial factors in time.

3.2.2 Exponentials with different exponents

$$(e^{-at}u(t)) * (e^{-bt}u(t)) = \frac{e^{-bt} - e^{-at}}{a - b}u(t), \quad a \neq b.$$

This pair is a compact exam lookup for cascaded first-order causal modes.

3.2.3 Causal exponential and step

$$(e^{-at}u(t)) * u(t) = \frac{1 - e^{-at}}{a}u(t), \quad a \neq 0.$$

This formula converts a first-order impulse response into a step response.

3.2.4 Causal exponential and ramp

$$(e^{-at}u(t)) * r(t) = \left(\frac{t}{a} - \frac{1}{a^2} + \frac{e^{-at}}{a^2} \right) u(t).$$

This formula converts a first-order impulse response into a ramp response.

3.2.5 Graphical convolution support

$$\text{supp}(f * g) = \text{supp}(f) + \text{supp}(g).$$

The output is nonzero only where the shifted signals overlap. For finite pulses, the breakpoints are sums of the original interval endpoints.

4 Fourier Series

4.1 Periodic Signal Representation

4.1.1 Exponential Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}, \quad D_n = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jn\omega_0 t} dt.$$

The exponential Fourier series expands a periodic signal in orthogonal complex exponentials.

4.1.2 Trigonometric Fourier series

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t).$$

The trigonometric series is useful for real periodic signals and separates cosine and sine content.

4.1.3 Relation between exponential and trigonometric coefficients

$$D_0 = a_0, \quad D_n = \frac{a_n - jb_n}{2}, \quad D_{-n} = \frac{a_n + jb_n}{2}.$$

For real signals, $D_{-n} = D_n^*$. Magnitude and phase follow from the complex coefficients.

4.1.4 Amplitude-phase Fourier series

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n), \quad C_n = 2|D_n|, \quad \theta_n = \angle D_n.$$

The amplitude-phase form describes the spectrum by harmonic amplitudes and phases.

4.2 Symmetry

4.2.1 Fourier series of real even signals

$$x(t) = x(-t) \Rightarrow D_n \in \mathbb{R}, \quad b_n = 0.$$

Real even periodic signals contain only cosine terms in the trigonometric form.

4.2.2 Fourier series of real odd signals

$$x(t) = -x(-t) \Rightarrow D_n \in j\mathbb{R}, \quad a_n = 0.$$

Real odd periodic signals contain only sine terms in the trigonometric form.

4.2.3 Parseval relation for Fourier series

$$\frac{1}{T_0} \int_{t_0}^{t_0+T_0} |x(t)|^2 dt = \sum_{n=-\infty}^{\infty} |D_n|^2.$$

Average signal power equals the sum of powers in all Fourier series coefficients.

4.2.4 Aperiodic time window reconstructed from Fourier series

$$x_p(t) = \sum_{k=-\infty}^{\infty} x_a(t - kT_0).$$

A Fourier series computed from a finite time window represents the periodic extension of that window.

5 Fourier Transformation

5.1 Definition and Inverse

5.1.1 Continuous-time Fourier transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt.$$

The Fourier transform maps a continuous-time signal to angular frequency in rad/s.

5.1.2 Inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega.$$

The inverse transform reconstructs the time signal from the complete frequency spectrum.

5.1.3 Magnitude and phase spectrum

$$X(\omega) = |X(\omega)|e^{j\angle X(\omega)}.$$

Magnitude describes frequency content amplitude. Phase describes time alignment of the frequency components.

5.2 Fourier Transform Pairs

5.2.1 Impulse and constant

$$\delta(t) \longleftrightarrow 1, \quad 1 \longleftrightarrow 2\pi\delta(\omega).$$

An impulse contains equal weight at all frequencies. A constant signal has all spectral mass at zero frequency.

5.2.2 Complex exponential and sinusoid

$$\begin{aligned} e^{j\omega_0 t} &\longleftrightarrow 2\pi\delta(\omega - \omega_0), \\ \cos\omega_0 t &\longleftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]. \end{aligned}$$

Periodic sinusoids transform into impulses at their positive and negative angular frequencies.

5.2.3 Causal exponential

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a + j\omega}, \quad a > 0.$$

The one-sided decaying exponential produces a first-order frequency response.

5.2.4 Rectangular pulse and sinc spectrum

$$x(t) = \begin{cases} 1, & |t| < T/2, \\ 0, & |t| > T/2, \end{cases} \quad X(\omega) = \frac{2\sin(\omega T/2)}{\omega}.$$

A finite rectangular pulse has a sinc-shaped spectrum with zeros at multiples of $2\pi/T$.

5.3 Properties

5.3.1 Fourier transform linearity

$$ax_1(t) + bx_2(t) \longleftrightarrow aX_1(\omega) + bX_2(\omega).$$

Linearity allows spectra to be built from simpler transform pairs.

5.3.2 Fourier transform time shift

$$x(t - t_0) \longleftrightarrow e^{-j\omega t_0} X(\omega).$$

A time delay adds a linear phase term and does not change the magnitude spectrum.

5.3.3 Fourier transform frequency shift

$$e^{j\omega_0 t} x(t) \longleftrightarrow X(\omega - \omega_0).$$

Multiplication by a complex sinusoid shifts the spectrum in frequency.

5.3.4 Fourier transform differentiation

$$\frac{d^n x(t)}{dt^n} \longleftrightarrow (j\omega)^n X(\omega).$$

Differentiation in time multiplies the spectrum by powers of $j\omega$.

5.3.5 Fourier transform convolution property

$$x(t) * h(t) \longleftrightarrow X(\omega)H(\omega).$$

Convolution in time becomes multiplication in the Fourier domain.

5.3.6 Fourier transform symmetry for real signals

$$x(t) \in \mathbb{R} \Rightarrow X(-\omega) = X^*(\omega).$$

Real signals have conjugate-symmetric spectra. The magnitude is even and the phase is odd where the phase is defined.

5.3.7 Fourier transform symmetry for even and odd signals

$$\begin{aligned} x(t) \text{ real and even} &\Rightarrow X(\omega) \text{ real and even,} \\ x(t) \text{ real and odd} &\Rightarrow X(\omega) \text{ imaginary and odd.} \end{aligned}$$

Signal symmetry strongly restricts the spectrum and is useful for true-false exam questions.

6 Sampling and Applications of Fourier Transformation

6.1 Sampling

6.1.1 Impulse-train sampling

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s).$$

Ideal sampling multiplies the signal by a periodic impulse train with sampling period T_s .

6.1.2 Sampling frequency and Nyquist frequency

$$f_s = \frac{1}{T_s}, \quad \omega_s = \frac{2\pi}{T_s}, \quad f_N = \frac{f_s}{2}, \quad \omega_N = \frac{\omega_s}{2}.$$

The Nyquist frequency is half the sampling frequency. It is the largest frequency that can be represented without aliasing for a bandlimited signal.

6.1.3 Spectrum of sampled signal

$$X_s(\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).$$

Sampling repeats the continuous-time spectrum at integer multiples of the sampling angular frequency.

6.1.4 Sampling theorem

$$\omega_s > 2\omega_{\max}.$$

A bandlimited signal with no content above $\omega_{\max}$ can be reconstructed from ideal samples if the sampling angular frequency exceeds twice the signal bandwidth.

6.1.5 Aliasing frequency

$$f_{\text{alias}} = |f - kf_s|, \quad \text{choose } k \in \mathbb{Z} \text{ so } 0 \leq f_{\text{alias}} \leq \frac{f_s}{2}.$$

Frequencies separated by integer multiples of f_s produce the same sampled sequence.

6.1.6 Ideal reconstruction

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right), \quad \operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}.$$

Ideal bandlimited reconstruction uses sinc interpolation from the sample values.

6.2 Fourier Applications

6.2.1 Amplitude modulation

$$x(t) \cos \omega_c t \longleftrightarrow \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)].$$

Multiplication by a carrier shifts copies of the baseband spectrum to $\pm\omega_c$.

6.2.2 System response in Fourier domain

$$Y(\omega) = H(j\omega)X(\omega).$$

For stable LTIC systems, the output spectrum is the input spectrum multiplied by the frequency response.

6.2.3 Magnitude in decibel

$$A_{\text{dB}} = 20 \log_{10} |H(j\omega)|.$$

Voltage and transfer-function magnitudes use $20 \log_{10}$. Power ratios use $10 \log_{10}$.

7 ADC Performance

7.1 Quantization

7.1.1 ADC least significant bit

$$\Delta = \frac{V_{\text{FS}}}{2^N}.$$

The quantization step is the voltage represented by one code transition for an N -bit converter.

7.1.2 Quantization error range

$$e_q = x_q - x, \quad -\frac{\Delta}{2} \leq e_q < \frac{\Delta}{2}.$$

For an ideal rounding quantizer, the quantization error is bounded by half an LSB.

7.1.3 Quantization noise power

$$\sigma_q^2 = \frac{\Delta^2}{12}.$$

The uniform-noise model is used when the quantization error is approximately uncorrelated with the input.

7.2 Dynamic Range and Resolution

7.2.1 Ideal ADC signal-to-quantization-noise ratio

$$\text{SQNR}_{\text{dB}} \approx 6.02N + 1.76 \text{ dB}$$

for a full-scale sine wave. Each additional bit improves the ideal SQNR by approximately 6 dB.

7.2.2 Effective number of bits

$$\text{ENOB} = \frac{\text{SINAD}_{\text{dB}} - 1.76}{6.02}.$$

ENOB converts measured distortion and noise performance into an equivalent ideal bit count.

7.2.3 Signal-to-noise-and-distortion ratio

$$\text{SINAD}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}} + P_{\text{distortion}}} \right).$$

SINAD includes both noise and harmonic distortion relative to the fundamental signal power.

7.2.4 Total harmonic distortion

$$\text{THD}_{\text{dB}} = 10 \log_{10} \left(\frac{\sum_{k=2}^K P_k}{P_1} \right).$$

THD compares harmonic power to the fundamental power.

7.2.5 Spurious-free dynamic range

$$\text{SFDR}_{\text{dB}} = 20 \log_{10} \left(\frac{A_1}{A_{\text{spur,max}}} \right).$$

SFDR is the amplitude ratio between the fundamental and the largest spur.

7.3 Sampling Limits

7.3.1 Aperture jitter noise

$$\text{SNR}_{\text{jitter,dB}} \approx -20 \log_{10}(2\pi f_{\text{in}} \sigma_t).$$

Sampling-time uncertainty creates amplitude error proportional to input frequency and jitter standard deviation.

7.3.2 Oversampling process gain

$$G_{\text{OSR,dB}} = 10 \log_{10}(\text{OSR}), \quad \text{OSR} = \frac{f_s}{2B}.$$

Oversampling spreads quantization noise over a wider band. Lowpass filtering to bandwidth B improves in-band SNR by the process gain.

8 Laplace Transform

8.1 Definitions

8.1.1 Unilateral Laplace transform

$$X(s) = \mathcal{L}\{x(t)\} = \int_{0^-}^{\infty} x(t)e^{-st} dt.$$

The exams state that Laplace transforms are unilateral unless otherwise specified. Initial conditions are included through derivative rules.

8.1.2 Inverse Laplace transform

$$x(t) = \mathcal{L}^{-1}\{X(s)\}.$$

Inverse transformation is normally performed by partial fractions and transform pairs.

8.1.3 Region of convergence

$$X(s) = \int x(t)e^{-st} dt \quad \text{converges for } s \in \text{ROC}.$$

For causal rational signals, the region of convergence lies to the right of the rightmost pole.

8.2 Transform Pairs

8.2.1 Step, impulse, and ramp Laplace transforms

$$\delta(t) \leftrightarrow 1, \quad u(t) \leftrightarrow \frac{1}{s}, \quad tu(t) \leftrightarrow \frac{1}{s^2}.$$

These pairs are the basic inputs for impulse, step, and ramp responses.

8.2.2 Exponential Laplace transform

$$e^{-at}u(t) \leftrightarrow \frac{1}{s+a}.$$

First-order poles produce causal exponentials in time.

8.2.3 Damped sinusoid Laplace transforms

$$e^{-\alpha t} \cos \omega_d t u(t) \leftrightarrow \frac{s + \alpha}{(s + \alpha)^2 + \omega_d^2},$$
$$e^{-\alpha t} \sin \omega_d t u(t) \leftrightarrow \frac{\omega_d}{(s + \alpha)^2 + \omega_d^2}.$$

Complex conjugate poles produce exponentially damped sinusoids.

8.3 Laplace Properties

8.3.1 Linearity of Laplace transform

$$\mathcal{L}\{ax(t) + by(t)\} = aX(s) + bY(s).$$

Linearity allows transform expressions to be assembled term by term.

8.3.2 Laplace derivative rule

$$\mathcal{L}\left\{\frac{dx}{dt}\right\} = sX(s) - x(0^-),$$
$$\mathcal{L}\left\{\frac{d^2x}{dt^2}\right\} = s^2X(s) - sx(0^-) - \dot{x}(0^-).$$

Unilateral derivative rules insert initial conditions directly into algebraic equations.

8.3.3 Laplace integration rule

$$\mathcal{L}\left\{\int_{0^-}^t x(\tau) d\tau\right\} = \frac{X(s)}{s}.$$

Time integration corresponds to division by s .

8.3.4 Laplace time shift

$$x(t - t_0)u(t - t_0) \leftrightarrow e^{-st_0}X(s), \quad t_0 > 0.$$

A causal delay multiplies the Laplace transform by an exponential factor.

8.3.5 Laplace convolution property

$$x(t) * h(t) \leftrightarrow X(s)H(s).$$

Convolution in time is multiplication in the Laplace domain.

8.3.6 Initial value theorem

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s).$$

The theorem applies when the limit exists and no impulse at the origin invalidates the ordinary value.

8.3.7 Final value theorem

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s).$$

The theorem applies when all poles of $sX(s)$ are in the open left half-plane, except possibly a simple pole at the origin in $X(s)$.

9 Applications of Laplace

9.1 Transfer Functions

9.1.1 Transfer function from differential equation

$$\left(\sum_{k=0}^N a_k s^k \right) Y(s) = \left(\sum_{m=0}^M b_m s^m \right) X(s), \quad H(s) = \frac{Y(s)}{X(s)} = \frac{P(s)}{Q(s)}.$$

With zero initial conditions, derivatives are replaced by powers of s . The rational transfer function describes the zero-state system.

9.1.2 Frequency response from transfer function

$$H(j\omega) = H(s)|_{s=j\omega}.$$

For stable systems, the Fourier-domain transfer function is obtained by evaluating the Laplace transfer function on the imaginary axis.

9.1.3 Poles and zeros

$$H(s) = K \frac{\prod_i (s - z_i)}{\prod_k (s - p_k)}.$$

Zeros z_i make the transfer function vanish. Poles p_k determine natural modes and stability.

9.1.4 Partial fraction expansion

$$\frac{B(s)}{\prod_{k=1}^N (s - p_k)} = \sum_{k=1}^N \frac{A_k}{s - p_k}, \quad A_k = (s - p_k)H(s)|_{s=p_k}.$$

For distinct poles, residues give the time-domain exponential weights.

9.2 Circuit Relations

9.2.1 Resistor, capacitor, and inductor laws

$$v_R(t) = Ri_R(t), \quad i_C(t) = C \frac{dv_C(t)}{dt}, \quad v_L(t) = L \frac{di_L(t)}{dt}.$$

These component equations convert circuits into differential equations in the time domain.

9.2.2 Laplace impedances

$$Z_R = R, \quad Z_C = \frac{1}{sC}, \quad Z_L = sL.$$

With zero initial conditions, circuit analysis in the Laplace domain uses algebraic impedances.

9.2.3 Voltage divider transfer function

$$H(s) = \frac{Y(s)}{X(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}.$$

Voltage dividers are a direct way to identify first-order RC and RL filter transfer functions.

9.2.4 First-order RC lowpass and highpass

$$H_{LP}(s) = \frac{1}{1 + sRC}, \quad H_{HP}(s) = \frac{sRC}{1 + sRC}.$$

The cutoff angular frequency is $\omega_c = 1/(RC)$. The lowpass passes DC, while the highpass rejects DC.

9.3 Response Calculation

9.3.1 Zero-state response in Laplace domain

$$Y_{zs}(s) = H(s)X(s).$$

The zero-state response is found by multiplying input transform and transfer function, then inverting.

9.3.2 Zero-input response from initial conditions

$$Q(s)Y(s) - \text{initial-condition terms} = 0.$$

The zero-input response is found by setting the input to zero and retaining the unilateral derivative initial-condition terms.

9.3.3 Step response final value

$$y_\infty = \lim_{s \rightarrow 0} sH(s) \frac{1}{s} = H(0).$$

For a stable system, the final value of the unit step response equals the DC gain.

10 Second-Order Systems and Bode Plot

10.1 Second-Order Standard Form

10.1.1 Second-order denominator

$$Q(s) = s^2 + 2\zeta\omega_n s + \omega_n^2.$$

ω_n is the natural angular frequency and ζ is the damping factor.

10.1.2 Second-order poles

$$p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}.$$

For $0 < \zeta < 1$, the poles are complex conjugates with damped angular frequency $\omega_d = \omega_n\sqrt{1 - \zeta^2}$.

10.1.3 Damping cases

ζ	case
$0 < \zeta < 1$	underdamped
$\zeta = 1$	critically damped
$\zeta > 1$	overdamped

The damping factor determines whether the natural response oscillates, has repeated real poles, or has distinct real poles.

10.1.4 Quality factor

$$Q = \frac{1}{2\zeta}.$$

The quality factor is large for lightly damped second-order systems and small for strongly damped systems.

10.1.5 Second-order lowpass, highpass, and bandpass

$$H_{\text{LP}}(s) = K \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

$$H_{\text{HP}}(s) = K \frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad H_{\text{BP}}(s) = K \frac{2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

The numerator determines whether low, high, or middle frequencies are passed.

10.2 Bode Plot

10.2.1 Bode magnitude and phase

$$M_{\text{dB}}(\omega) = 20 \log_{10} |H(j\omega)|, \quad \phi(\omega) = \arg H(j\omega).$$

Bode plots show transfer-function magnitude in decibel and phase versus logarithmic frequency.

10.2.2 Constant gain Bode rule

$$K \Rightarrow 20 \log_{10} |K| \text{ dB}, \quad \angle K = \begin{cases} 0^\circ, & K > 0, \\ 180^\circ, & K < 0. \end{cases}$$

Constant factors shift the magnitude plot without changing slope.

10.2.3 Real zero and real pole Bode rules

$$1 + \frac{s}{\omega_z} : +20 \text{ dB/dec after } \omega_z,$$

$$\frac{1}{1 + s/\omega_p} : -20 \text{ dB/dec after } \omega_p.$$

A first-order zero raises the asymptotic slope by 20 dB/decade. A first-order pole lowers it by 20 dB/decade.

10.2.4 Origin pole and origin zero Bode rules

$$s^m \Rightarrow +20m \text{ dB/dec}, \quad \frac{1}{s^m} \Rightarrow -20m \text{ dB/dec}.$$

Zeros or poles at the origin set the initial Bode magnitude slope and add constant phase of $\pm 90^\circ$ per order.

10.2.5 Second-order pole pair Bode rule

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Rightarrow -40 \text{ dB/dec after } \omega_n.$$

A second-order pole pair contributes a total phase transition of approximately -180° . Resonance becomes pronounced for small ζ .

10.2.6 Resonance peak

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}, \quad \omega_r = \omega_n \sqrt{1-2\zeta^2}, \quad 0 < \zeta < \frac{1}{\sqrt{2}}.$$

Underdamped lowpass systems have a resonance peak only for sufficiently small damping.

11 Bode Plot and Pole-Zero Filter Design

11.1 Pole-Zero Interpretation

11.1.1 Frequency response from pole-zero distances

$$|H(j\omega)| = |K| \frac{\prod_i |j\omega - z_i|}{\prod_k |j\omega - p_k|}, \quad \angle H(j\omega) = \angle K + \sum_i \angle(j\omega - z_i) - \sum_k \angle(j\omega - p_k).$$

Magnitude is the ratio of distances from $j\omega$ to zeros and poles. Phase is the sum of zero angles minus pole angles.

11.1.2 Filter type from zeros at the origin

$$H(s) \propto s^m \Rightarrow H(0) = 0.$$

Zeros at the origin reject DC and are characteristic of highpass and bandpass transfer functions.

11.1.3 Notch filter zeros

$$H_{\text{notch}}(s) = K \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}.$$

Zeros on the imaginary axis at $\pm j\omega_0$ reject the notch frequency.

11.2 Frequency Transformations

11.2.1 Lowpass frequency scaling

$$H_{\text{LP}}(s) = H_n\left(\frac{s}{\omega_c}\right).$$

Frequency scaling moves a normalized cutoff from 1 rad/s to ω_c .

11.2.2 Lowpass to highpass substitution

$$s_n = \frac{\omega_c}{s}.$$

Substituting $s_n = \omega_c/s$ in a normalized lowpass transfer function gives a highpass transfer function with cutoff ω_c .

11.2.3 Lowpass to bandpass substitution

$$s_n = \frac{s^2 + \omega_0^2}{Bs}, \quad B = \omega_2 - \omega_1, \quad \omega_0 = \sqrt{\omega_1\omega_2}.$$

The lowpass-to-bandpass transformation maps a normalized prototype to passband edges ω_1 and ω_2 .

11.2.4 Cascade of filter stages

$$H(s) = \prod_{k=1}^M H_k(s).$$

Higher-order filters are implemented as cascades of first-order and second-order stages. Bode magnitudes in dB and phases add across stages.

11.3 Tracking Filters

11.3.1 Tracking-filter numerator choice

$$T(s) = \frac{as + b}{(s + p_1)(s + p_2)}.$$

The numerator coefficients place a zero and set the gain. Exam tasks use this form to match specified gain or phase behavior at selected frequencies.

11.3.2 Solving gain constraints

$$T(0) = \frac{b}{p_1 p_2}, \quad \lim_{s \rightarrow \infty} sT(s) = a.$$

Low-frequency and high-frequency asymptotes provide direct algebraic constraints on numerator coefficients.

12 Butterworth Filter Design and Sensitivity

12.1 Butterworth Prototypes

12.1.1 Butterworth magnitude response

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}}.$$

A Butterworth filter is maximally flat in the passband. At $\omega = \omega_c$, the magnitude is $1/\sqrt{2}$, or -3 dB.

12.1.2 Butterworth pole angles

$$p_k = \omega_c e^{j\left(\frac{\pi}{2} + \frac{(2k-1)\pi}{2N}\right)}, \quad k = 1, \dots, N.$$

The stable Butterworth poles lie on a circle of radius ω_c in the left half-plane.

12.1.3 Butterworth second-order stage

$$H_k(s) = \frac{\omega_c^2}{s^2 + a_k \omega_c s + \omega_c^2}, \quad a_k = 2\zeta_k.$$

Even-order Butterworth filters are cascades of second-order sections. The stage coefficient a_k sets damping and quality factor.

12.1.4 Common Butterworth coefficients

N	second-order a_k values
2	1.4142
4	0.7654, 1.8478
6	0.5176, 1.4142, 1.9319
8	0.3902, 1.1111, 1.6629, 1.9616

These coefficients are used when decomposing normalized Butterworth filters into cascaded second-order stages.

12.2 Sallen-Key Lowpass

12.2.1 Sallen-Key lowpass transfer function

$$H_{\text{SK,LP}}(s) = \frac{K}{s^2 R_1 R_2 C_1 C_2 + s(R_1 C_1 + R_1 C_2 + R_2 C_2(1 - K)) + 1}.$$

The Sallen-Key lowpass realizes a second-order denominator using two resistors, two capacitors, and an amplifier gain K .

12.2.2 Sallen-Key natural frequency and damping coefficient

$$\omega_n = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}},$$

$$a = 2\zeta = \frac{R_1 C_1 + R_1 C_2 + R_2 C_2(1 - K)}{\sqrt{R_1 R_2 C_1 C_2}}.$$

Component products set the natural frequency. Component ratios and gain set damping.

12.2.3 Unity-gain Sallen-Key lowpass

$$K = 1 \quad \Rightarrow \quad a = \frac{R_1 C_1 + R_1 C_2}{\sqrt{R_1 R_2 C_1 C_2}}.$$

Unity-gain design removes the $R_2 C_2(1 - K)$ term and is used in the low-sensitivity design slides.

12.3 Sensitivity

12.3.1 Relative sensitivity

$$S_x^y = \frac{\partial y / y}{\partial x / x} = \frac{x}{y} \frac{\partial y}{\partial x}.$$

Sensitivity measures fractional output change per fractional component change.

12.3.2 Sensitivity of products and powers

$$y = K x_1^{\alpha_1} x_2^{\alpha_2} \Rightarrow S_{x_i}^y = \alpha_i.$$

Power-law expressions give sensitivities directly from exponents.

12.3.3 Monte Carlo component tolerance

$$x_i = x_{i,\text{nom}}(1 + \epsilon_i).$$

Monte Carlo analysis perturbs component values within their tolerance ranges to estimate spread in cutoff frequency, gain, and damping.

12.4 Scaling

12.4.1 Frequency scaling of RC filters

$$\omega'_n = K_F \omega_n, \quad R' C' = \frac{RC}{K_F}.$$

Increasing cutoff frequency requires reducing the relevant RC products.

12.4.2 Impedance scaling

$$R' = K_Z R, \quad C' = \frac{C}{K_Z}.$$

Impedance scaling changes practical component values while preserving the same RC products and transfer function.

13 Butterworth Highpass and AC-Coupled In-Amp

13.1 Butterworth Highpass

13.1.1 Second-order Butterworth highpass stage

$$H_{\text{HP}}(s) = K \frac{s^2}{s^2 + a\omega_c s + \omega_c^2}.$$

The highpass numerator places two zeros at the origin. The denominator uses the same Butterworth damping coefficients as the lowpass prototype.

13.1.2 Lowpass-to-highpass Butterworth transformation

$$H_{\text{HP}}(s) = H_{\text{LP},n}\left(\frac{\omega_c}{s}\right).$$

This substitution maps the normalized Butterworth lowpass prototype to a highpass cutoff at ω_c .

13.1.3 Highpass cutoff conversion

$$\omega_c = 2\pi f_c.$$

Filter design often specifies cutoff frequency in hertz, while transfer functions use angular frequency in rad/s.

13.2 AC Coupling

13.2.1 Single-pole AC-coupling highpass

$$H_{AC}(s) = \frac{sRC}{1 + sRC} = \frac{s}{s + \omega_c}, \quad \omega_c = \frac{1}{RC}.$$

AC coupling rejects electrode offsets and slow baseline drift while passing higher-frequency signal components.

13.2.2 Cascaded AC-coupling stages

$$H_{\text{total}}(s) = \prod_{k=1}^M \frac{s}{s + \omega_{c,k}}.$$

Multiple coupling stages increase low-frequency attenuation and add phase shift near each cutoff.

13.3 Instrumentation Amplifier

13.3.1 Differential and common-mode signals

$$v_d = v_2 - v_1, \quad v_{\text{cm}} = \frac{v_1 + v_2}{2}.$$

Biopotential front ends separate the desired differential signal from common-mode interference.

13.3.2 Differential gain and common-mode gain

$$v_o = A_d v_d + A_{\text{cm}} v_{\text{cm}}.$$

An ideal instrumentation amplifier has large differential gain and zero common-mode gain.

13.3.3 Common-mode rejection ratio

$$\text{CMRR} = \left| \frac{A_d}{A_{\text{cm}}} \right|, \quad \text{CMRR}_{\text{dB}} = 20 \log_{10} \left| \frac{A_d}{A_{\text{cm}}} \right|.$$

CMRR quantifies rejection of common-mode signals relative to differential signals.

13.3.4 Three-op-amp instrumentation amplifier gain

$$A_d = \left(1 + \frac{2R}{R_G} \right) \frac{R_3}{R_2}.$$

The gain is set by the input-stage gain resistor R_G and the output difference-amplifier resistor ratio.

13.3.5 Resistor mismatch and common-mode leakage

$$\frac{R_3}{R_2} \neq \frac{R_5}{R_4} \Rightarrow A_{\text{cm}} \neq 0.$$

Balanced resistor ratios are required for common-mode cancellation. Mismatch degrades CMRR.

14 Exam Required Extra Topics

14.1 Exam Formula Sheet Conventions

14.1.1 One-sided initial conditions

$$x(0^-), \quad \dot{x}(0^-), \quad y(0^-), \quad \dot{y}(0^-).$$

Exam problems use unilateral Laplace transforms and left-limit initial conditions. These values enter the derivative rules before solving for $Y(s)$.

14.1.2 Determining whether an exam statement is false

one false clause in an answer option $\Rightarrow$ the option is false.

Several multiple-choice exams use answer options containing multiple claims. Every claim in an option must be true for the option to be true.

14.2 Distributional Frequency Responses

14.2.1 Undamped oscillator frequency response

$$\mathcal{F}\{\sin \omega_0 t u(t)\} = \text{PV} \frac{\omega_0}{\omega_0^2 - \omega^2} \frac{\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].$$

Marginally stable systems can produce Fourier-domain expressions with principal-value and impulse terms.

14.2.2 Imaginary-axis poles and ordinary frequency response

$$H(s) = \frac{1}{s^2 + \omega_0^2} \quad \text{has poles at } s = \pm j\omega_0.$$

Substitution $s = j\omega$ is not an ordinary finite frequency response at pole frequencies on the imaginary axis.